

## **Solve the following problems using divide and conquer:**

1. Count total even numbers in an array..
2. Count total vowels in a string.
3. Find the largest pair in a given array.
4. Given an array of size  $n$ , return the majority element. The majority element is the element that appears more than  $\lfloor n/2 \rfloor$  times. You may assume that the majority element always exists in the array.
5. You have an array of integers where some elements may be negative, zero, or positive. Your goal is to find the contiguous subarray (subarray with consecutive elements) with the maximum sum of non-negative numbers. (Note that, the subarray shouldn't have any negative number.)
6. You are given a list of students who have a name, id and cgpa. Your task is to find the student with the highest cgpa. [you can't sort the list]
7. You are given a sequence of  $n$  integers. For each position  $i$  ( $0 \leq i < n$ ), determine how many elements to the right of that position are strictly smaller than the element at position  $i$ .